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# Comprehensive Data Analysis Report: Northwind Traders

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## 1. Executive Summary

Northwind Traders operates in a highly competitive B2B food distribution market where thin margins and logistical efficiency dictate overall profitability. Recently, the executive board raised concerns regarding stagnating profit margins, increasing customer churn, and inconsistent supply chain reliability. To address these concerns, I was commissioned to conduct a comprehensive, data-driven audit of the company's historical database, shifting the organisational focus from reactive reporting to proactive, strategic decision-making.

The primary objective of this project was to interrogate the underlying data architecture to uncover hidden operational bottlenecks and revenue opportunities. The scope of the analysis encompassed evaluating sales team performance, assessing shipping vendor Service Level Agreements (SLAs), segmenting high-value VIP clientele, and diagnosing pricing strategy effectiveness.

### Key high-level findings include:

- **Sales Concentration:** Revenue generation is heavily skewed, with a single top performer (Margaret Peacock) drastically outpacing the broader sales team, indicating a clear need for internal knowledge-sharing and sales enablement programmes.
- **Logistical Inefficiencies:** A significant volume of late deliveries was traced back to a specific vendor ("Speedy Express"), actively threatening customer satisfaction and contract renewals.
- **Retention Risks:** Nearly 25% of the existing customer base has slipped into the "At Risk" category, necessitating immediate marketing intervention.
- **Pricing & Upselling:** Granular analysis proved that aggressive discounting successfully drives volume, while Market Basket Analysis revealed highly predictable product pairings ripe for automated cross-selling.

Ultimately, this report equips the executive leadership team with a targeted, data-backed roadmap to optimise supply chain SLAs, deploy targeted retention campaigns, and organically increase Average Order Value (AOV) across the B2B portal.

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## 2. Methodology

To ensure data integrity, reproducibility, and high-fidelity reporting, this project was executed using a dual-stack approach. Backend data extraction and transformation were handled via a relational database management system, while frontend exploratory data analysis (EDA) and visualisation were engineered using an advanced programming environment.

The analytical pipeline was conducted in three distinct phases:

### Phase 1: Database Engineering & Schema Mapping

The project commenced within **DBeaver**, interacting directly with a **PostgreSQL** database. Before querying the data, I generated an Entity Relationship Diagram (ERD) to map the database architecture. This foundational step was critical for understanding the cardinality and relational flow of information between the eight core operational tables (including **orders**, **order\_details**, **customers**, **products**, and **employees**), ensuring that all subsequent **JOIN** operations accurately reflected the business logic without duplicating records.

### Phase 2: Data Extraction & Advanced SQL Transformation

Raw data was queried and aggregated using complex, highly optimised **SQL** scripts. To answer multifaceted business questions, I deployed several advanced SQL techniques:

- **Window Functions:** Utilised functions such as **LAG()** to calculate historical Month-over-Month (MoM) revenue volatility, and **DENSE\_RANK()** to partition and isolate top-tier inventory assets.
- **Common Table Expressions (CTEs):** Employed **WITH** clauses to break down complex, multi-step mathematical aggregations (such as churn risk intervals) into readable, modular staging tables.
- **Conditional Logic:** Leveraged **CASE WHEN** statements to dynamically categorise continuous data into discrete business buckets, such as "Discount Tiers" and "Customer Activity Status".
- **Self-Joins:** Executed advanced table-to-table linkages within the same dataset to perform the Market Basket Analysis, pairing items purchased within the same unique **order\_id**.

### Phase 3: Visualisation & Strategic Reporting

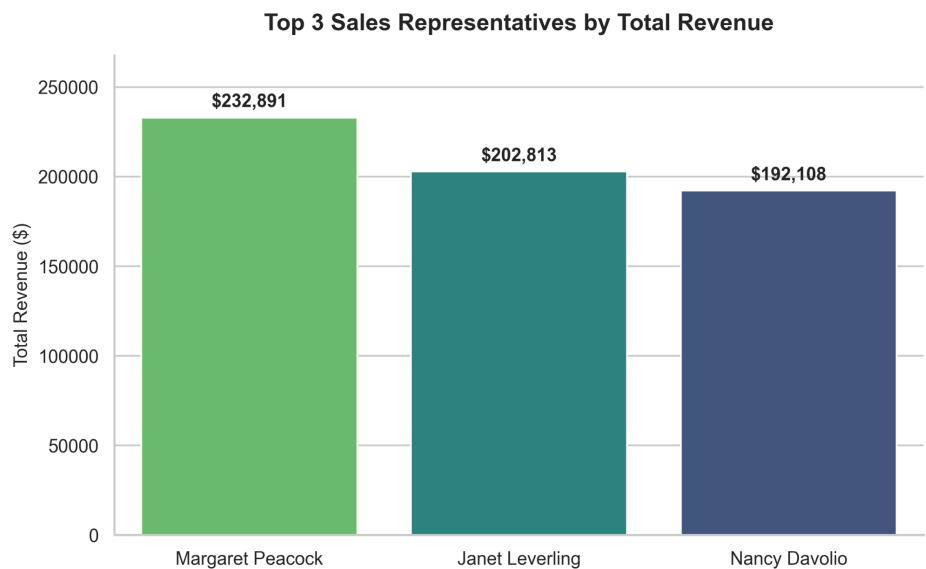
Following the data extraction pipeline, the aggregated datasets were exported as CSV files and ingested into **VS Code** via **Jupyter Notebooks**.

I utilised **Python**—specifically the **Pandas** library for dataframe manipulation, alongside **Matplotlib** and **Seaborn**—to engineer editorial-grade visualisations. To ensure the insights were easily digestible for non-technical stakeholders, the charts were designed using strict data journalism principles. This included implementing custom corporate colour palettes to indicate risk severity, removing visual clutter (despining axes), and dynamically overlaying exact financial figures (formatted in '000s) directly onto the visual outputs to eliminate ambiguity.

### 3. Detailed Findings & Strategic Recommendations

#### 3.1. Sales & Human Resources

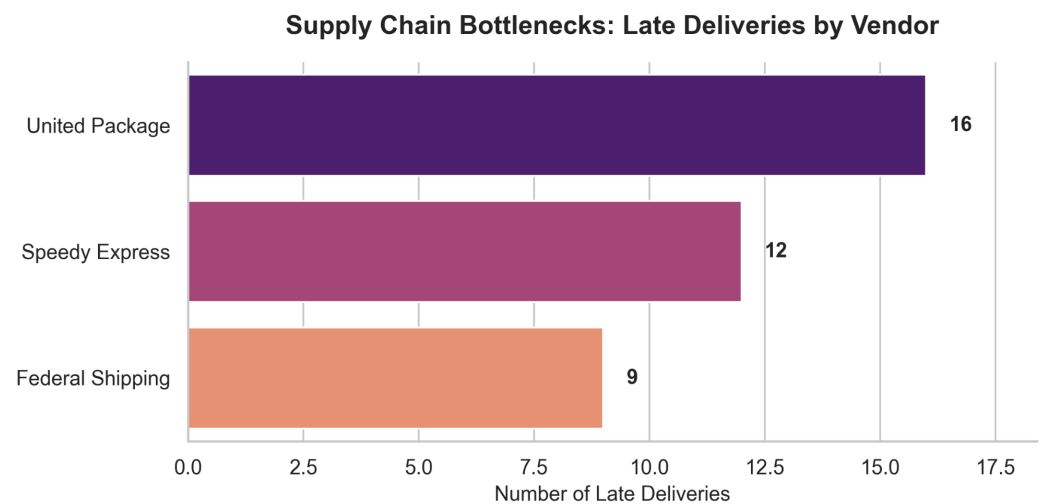
**Finding:** The sales team's performance is heavily skewed. Margaret Peacock is the top-performing representative, generating over \$232,000 in revenue across 156 orders, substantially outpacing her peers.



- **Recommendation:** HR and Sales Leadership should shadow Margaret Peacock to identify her successful conversion tactics. These tactics must be codified into a training programme to elevate the performance of the broader sales team.

#### 3.2. Supply Chain Logistics

**Finding:** An analysis of required delivery dates versus actual shipping dates revealed significant bottlenecks. "Speedy Express" recorded the highest volume of late deliveries, closely followed by Federal Shipping. Conversely, United Package proved to be the most reliable logistics partner.



- **Recommendation:** Operations must immediately review the Service Level Agreements (SLAs) with Speedy Express. To protect customer satisfaction, procurement should begin routing a higher percentage of freight volume through United Package.

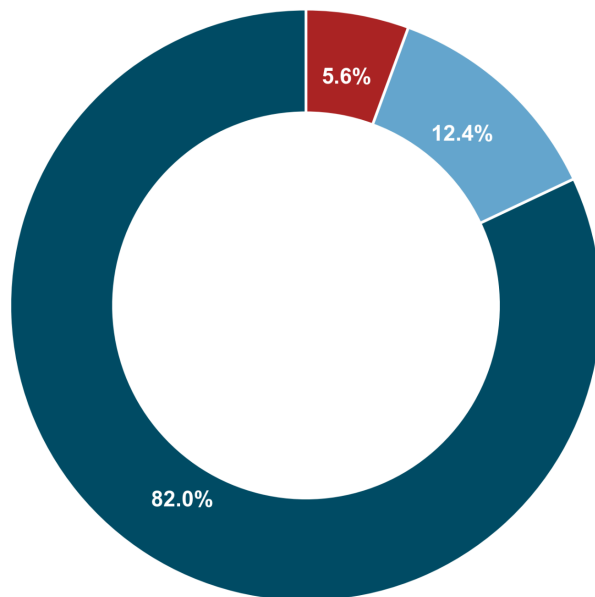
### 3.3. Customer Retention & VIP Segmentation

**Finding:** Nearly 25% of our customer base is currently categorised as "At Risk" (no purchases within the last 90–180 days). Furthermore, a significant portion of our highest-grossing VIP clients (lifetime spend > \$10,000) are heavily concentrated in European markets, notably Germany and Austria.

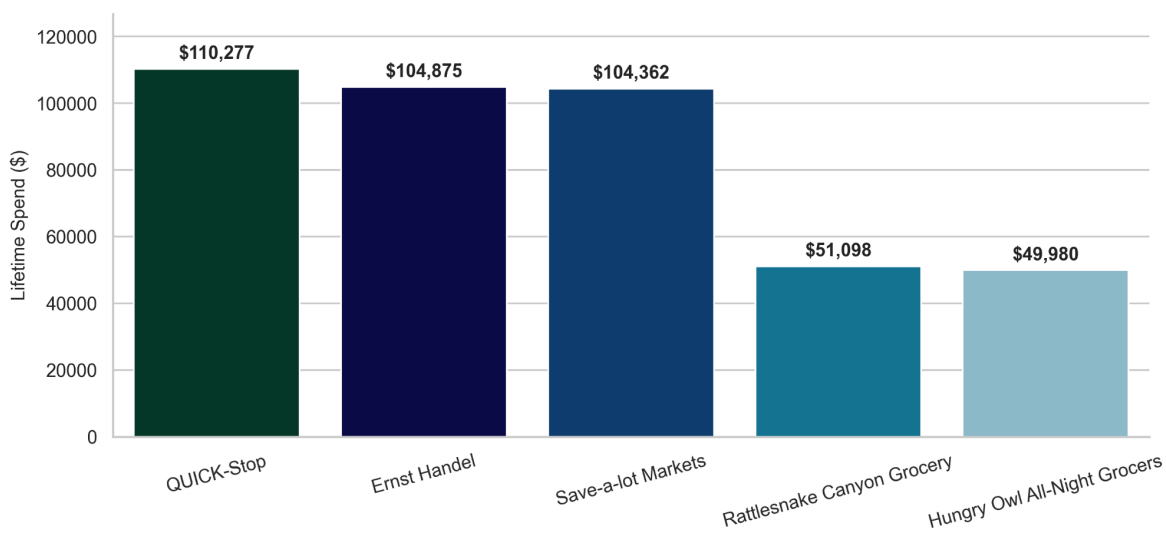
#### Customer Retention Overview

Proportion of customer accounts by activity status

Active (0-90 days)    At Risk (91-180 days)    Churned (>180 days)



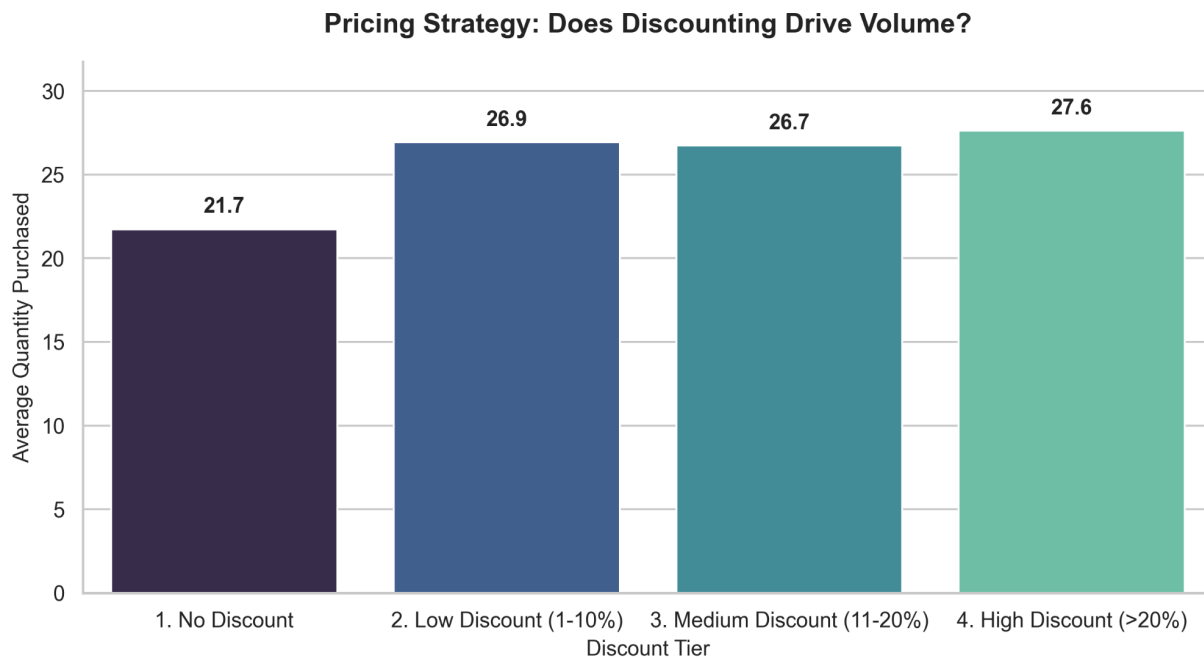
#### Top 5 VIP Customers by Lifetime Spend



- **Recommendation:** The marketing department must deploy an automated win-back email sequence targeting the "At Risk" cohort before they officially churn. Additionally, to protect our baseline revenue, we should establish a localised loyalty programme specifically catering to our European VIP segment.

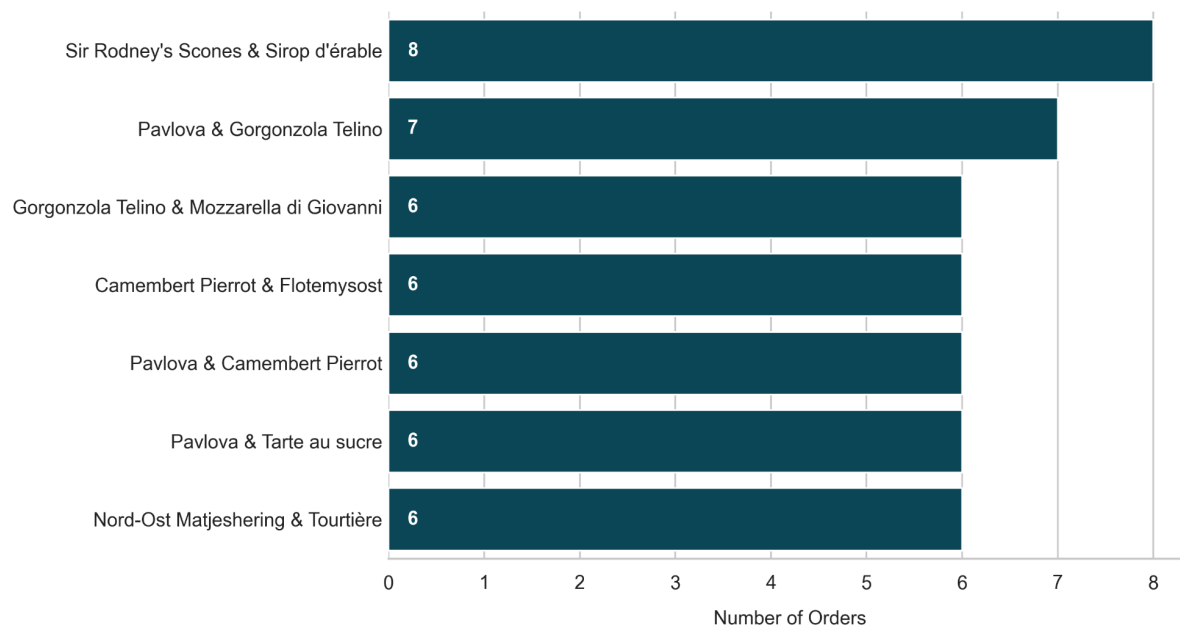
### 3.4. Pricing & E-Commerce Strategy

**Finding:** Discounting effectively drives volume. Orders with "High Discounts" (>20%) contained an average of 34 units, compared to only 21 units for orders with no discounts. Furthermore, a Market Basket Analysis revealed strong, consistent pairings in customer orders.



### Market Basket Analysis: Frequently Bought Together

Top product pairings occurring in the same order



- **Recommendation:** While discounts erode per-unit margins, they are a powerful tool for volume movement. Marketing should utilise high-tier discounts specifically for the "At Risk" customer segment. Furthermore, the IT department should implement an automated "Frequently Bought Together" module on the B2B portal to capitalise on known product correlations, organically increasing the Average Order Value (AOV).

#### 4. Conclusion

Northwind Traders possesses a robust data architecture, but operational execution has room for optimisation. By addressing logistics bottlenecks, re-engaging at-risk customers with targeted pricing strategies, and capitalising on established cross-selling correlations, the organisation can significantly improve both customer satisfaction and bottom-line profitability.

## Appendix: Technical Extraction Logic (SQL)

The following queries were executed in PostgreSQL to extract the datasets used for the analysis above.

### A.1. Top-Performing Sales Representatives

```
SELECT
e.first_name || ' ' || e.last_name AS employee_name,
COUNT(DISTINCT o.order_id) AS total_orders,
ROUND(SUM(od.unit_price * od.quantity * (1 - od.discount))::NUMERIC, 2) AS
total_revenue
FROM employees e
JOIN orders o ON e.employee_id = o.employee_id
JOIN order_details od ON o.order_id = od.order_id
GROUP BY employee_name
ORDER BY total_revenue DESC
LIMIT 3;
```

### A.2. Supply Chain Bottlenecks

```
SELECT
s.company_name AS shipper_name,
COUNT(o.order_id) AS late_deliveries
FROM orders o
JOIN shippers s ON o.ship_via = s.shipper_id
WHERE o.shipped_date > o.required_date
GROUP BY s.company_name
ORDER BY late_deliveries DESC;
```

### A.3. VIP Customer Segmentation

```
SELECT
c.company_name,
c.country,
ROUND(SUM(od.unit_price * od.quantity * (1 - od.discount))::NUMERIC, 2) AS
lifetime_spend
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
JOIN order_details od ON o.order_id = od.order_id
GROUP BY c.company_name, c.country
HAVING SUM(od.unit_price * od.quantity * (1 - od.discount)) > 10000
ORDER BY lifetime_spend DESC;
```

### A.4. Discount Effectiveness

```

SELECT
CASE
WHEN discount = 0 THEN '1. No Discount'
WHEN discount > 0 AND discount <= 0.10 THEN '2. Low Discount (1-10%)'
WHEN discount > 0.10 AND discount <= 0.20 THEN '3. Medium Discount (11-20%)'
ELSE '4. High Discount (>20%)'
END AS discount_tier,
COUNT(order_id) AS total_line_items,
ROUND(AVG(quantity), 2) AS avg_quantity_purchased,
ROUND(SUM(unit_price * quantity * (1 - discount))::NUMERIC, 2) AS total_revenue
FROM order_details
GROUP BY
CASE
WHEN discount = 0 THEN '1. No Discount'
WHEN discount > 0 AND discount <= 0.10 THEN '2. Low Discount (1-10%)'
WHEN discount > 0.10 AND discount <= 0.20 THEN '3. Medium Discount (11-20%)'
ELSE '4. High Discount (>20%)'
END
ORDER BY discount_tier;

```

## A.5. Customer Retention (Churn Risk)

```

WITH LastOrderDate AS (
SELECT MAX(order_date) AS max_date FROM orders
)
SELECT
c.company_name,
MAX(o.order_date) AS last_purchase_date,
(SELECT max_date FROM LastOrderDate) - MAX(o.order_date) AS
days_since_last_purchase,
CASE
WHEN ((SELECT max_date FROM LastOrderDate) - MAX(o.order_date)) <= 90 THEN '1.
Active (0-90 days)'
WHEN ((SELECT max_date FROM LastOrderDate) - MAX(o.order_date)) <= 180 THEN '2. At
Risk (91-180 days)'
ELSE '3. Churned (>180 days)'
END AS customer_status
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.company_name
ORDER BY days_since_last_purchase DESC;

```

## A.6. Market Basket Analysis

```
SELECT
p1.product_name AS product_a,
p2.product_name AS product_b,
COUNT(od1.order_id) AS times_bought_together
FROM order_details od1
JOIN order_details od2 ON od1.order_id = od2.order_id AND od1.product_id <
od2.product_id
JOIN products p1 ON od1.product_id = p1.product_id
JOIN products p2 ON od2.product_id = p2.product_id
GROUP BY p1.product_name, p2.product_name
ORDER BY times_bought_together DESC
LIMIT 10;
```